CRITICAL PATH METHOD

Forty-sixth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



What is CPM?

- CPM or Critical Path Method is a statistical technique of project management in which planning, scheduling, organizing, coordination and control of **well-defined activities** are done.
- Developed in the late 1950s by Morgan R. Walker of DuPont and James E. Kelley Jr of Ramington Rand, independently.
- Initially used for industrial and construction projects (most notable is World Trade Center twin towers in 1966 in New York).
- Later used in aerospace and defense, software development, research and product development, business and event planning etc.

Framework CPM

- CPM follows six basic steps:
- Define the project and prepare the work breakdown structure.
- Develop the relationships among the **activities**. Decide which activities must precede and which must follow others.
- Draw the **network** connecting all the activities.
- Assign time and/or cost estimates to each activity.
- Compute the *longest* time path through the network. This is called the **critical path**.
- Use the network to help plan, schedule, monitor, and control the project.

- CPM help answer questions about projects with several activities:
  - When will the entire project be completed?
  - What are the critical activities or tasks in the project? Which activities will delay the entire project if they are late?
  - Which are the noncritical activities—the ones that can run late without delaying the whole project's completion?
  - What is the probability that the project will be completed by a specific date? (PERT)

- CPM help answer questions about projects with thousands of activities:
  - At any particular date, is the project **on schedule**, **behind schedule**, or **ahead of schedule**?
  - On any given date, is the money spent **equal to**, **less than**, or **greater than** the budgeted amount?
  - Are there enough resources available to finish the project on time?
  - If the project is to be finished in a shorter amount of time, what is the best way to accomplish this goal at the least cost?

Network diagram and approaches

- The initial step in CPM is to divide the entire project into several activities in accordance with the work breakdown structure.
- There are two approaches for drawing a project network:
  - **activity on node (AON)** and **activity on arrow (AOA)**.
- **In AON convention, nodes designate activities. Under AOA, *arrows* represent activities**. Activities consume time and resources. The basic difference between AON and AOA is that the nodes in an AON diagram represent activities.
- In an AOA network, the **nodes** represent the starting and finishing times of an activity and are also called ***events***. Thus, nodes in AOA consume neither time nor resources.

Network Representation

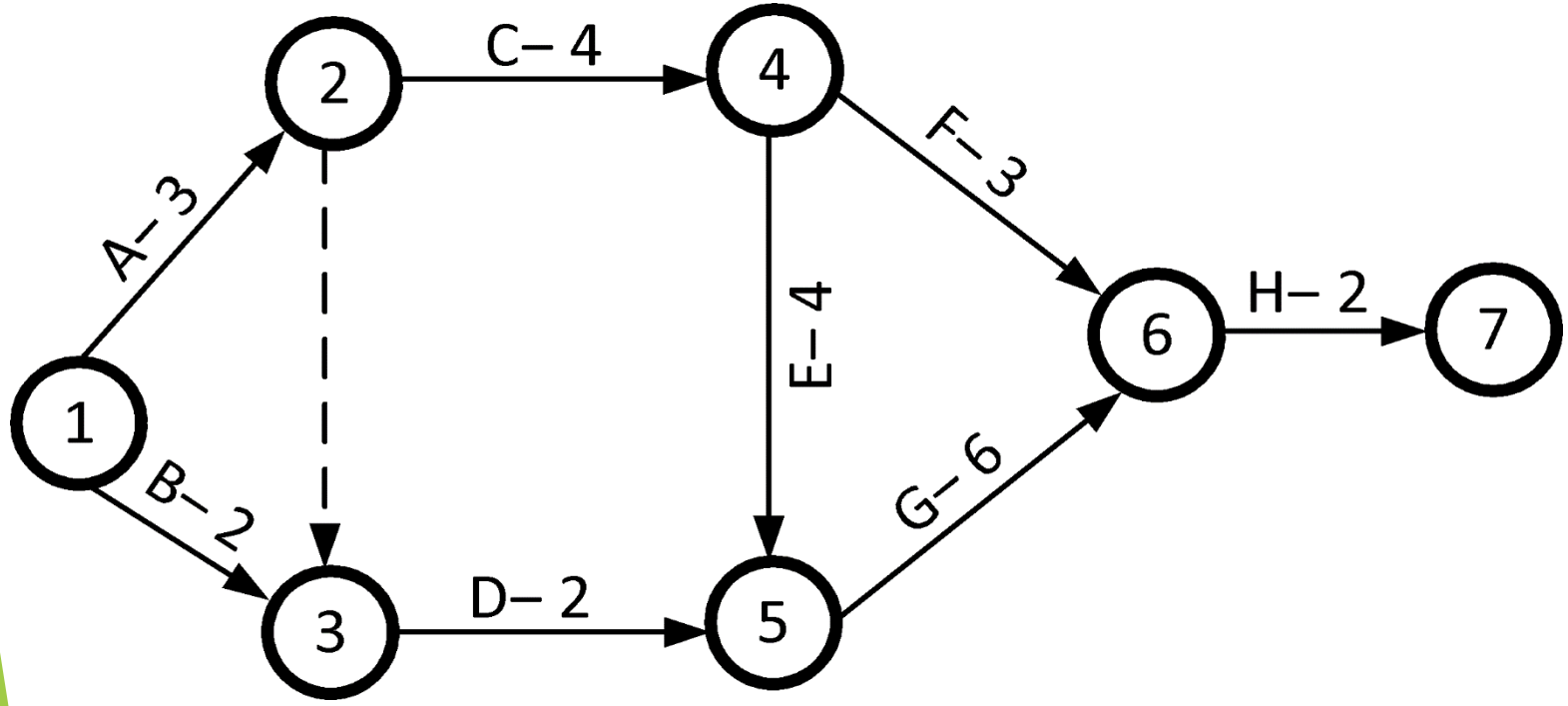
- Some rules for constructing a network are as follows:
- **Rule 1:** Each activity is represented by one and only arrow in the network.
- **Rule 2:** Each activity must be identified by two distinct end nodes.
- **Rule 3:** To minimize the correct precedence relationships, the following questions must be answered as each activity is added to the network.
  (a) What activities must immediately precede the current activity?
  (b) What activities must follow the current activity?
  (c) What activities must occur concurrently with the current analysis?

Example

- The activities involved and their predecessors for completing a project are shown in Table. The time duration (in weeks) required for each activity is also shown. Develop the associated network of the project.

Activities	Predecessor (s)	Duration (weeks)
A	--	3
B	--	2
C	A	4
D	A, B	2
E	C	4
F	C	3
G	D, E	6
H	F, G	2

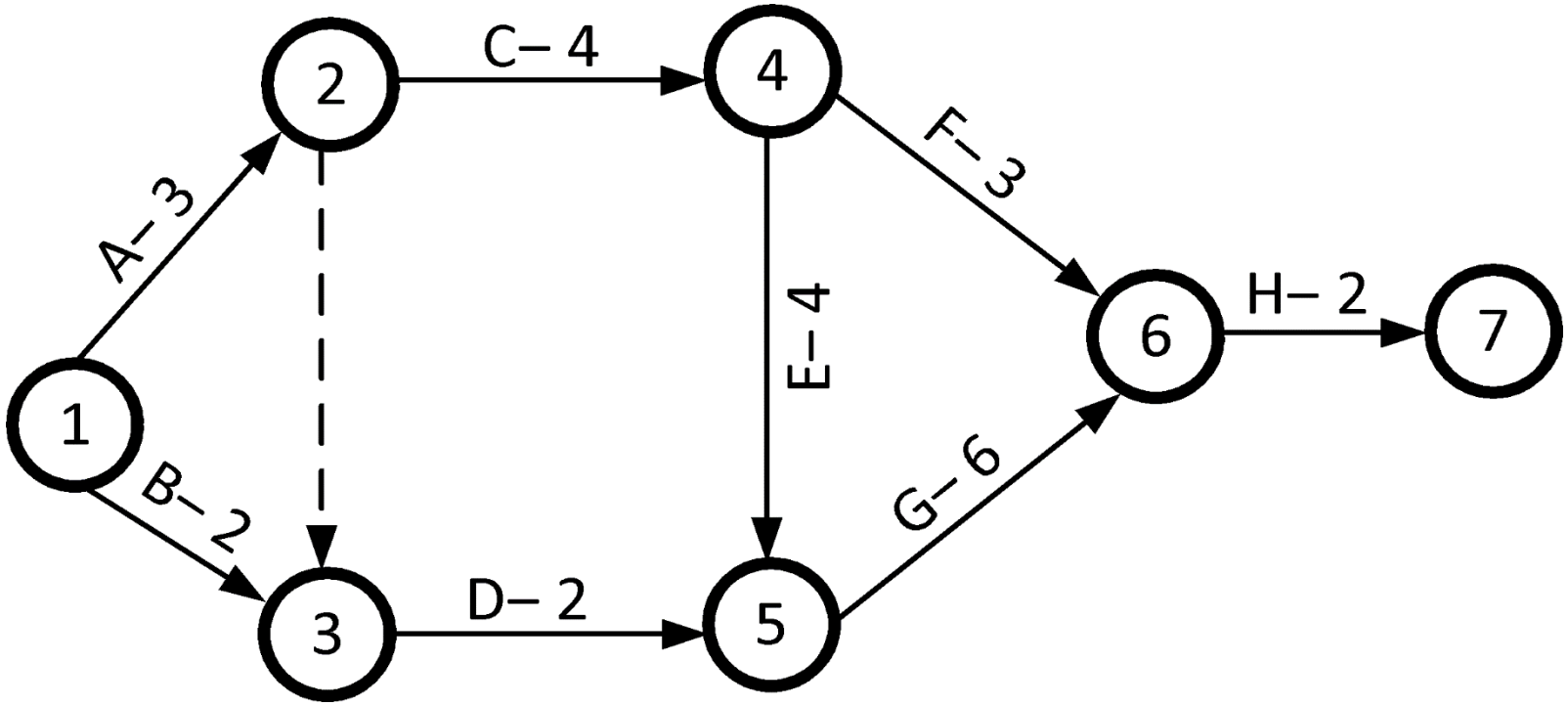
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Figure represents the network describing the relationships among different activities. The numbering of the nodes is done in a manner that indicates the direction of progress in the project.

Example



- **Dummy Activity:**
  - Dummy activity is a hypothetical activity which requires zero time and zero resources for completion. Dummy arrow represents an activity with zero duration.
  - The dummy activity (2,3) denoted by dotted arrow is used to produce unique end nodes for concurrent activities.

Determining the project schedule

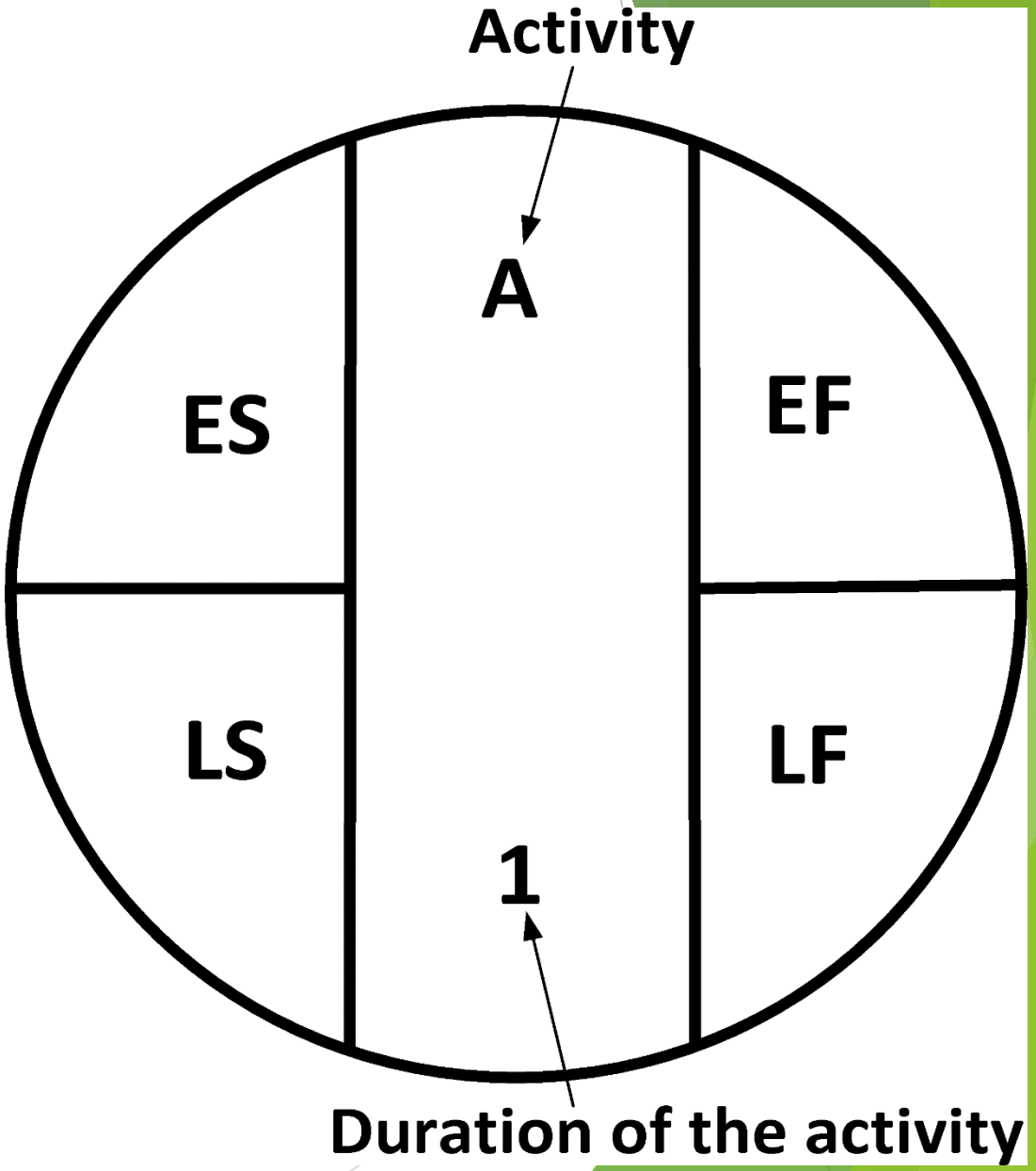
- To evaluate how long the project will take time to complete, the critical path is evaluated.
- **Critical Path:** The critical path is the longest path (in time) from Start to Finish. It indicates the minimum time necessary to complete the entire project.
- To find the critical path, it is necessary to calculate **two distinct starting and ending times for each activity**. These are defined as follows:
- **Earliest start (ES)**: earliest time at which an activity can start, assuming all predecessors have been completed.

Determining the project schedule

- **Earliest finish (EF)**: earliest time at which an activity can be finished.
- **Latest start (LS):** latest time at which an activity can start so as to not delay the completion time of the entire project
- **Latest finish (LF)**: latest time by which an activity has to finish so as to not delay the completion time of the entire project

Forward Pass

- The **earliest start (ES)** of an activity is shown in the top left corner of the node denoting that activity.
- The **earliest finish (EF)** is shown in the top right corner.
- The latest times, **latest start (LS)** and **latest finish (LF)**, are shown in the bottom-left and bottom-right corners, respectively.



Forward Pass

- **Earliest Start Time Rule:** Before an activity can start, all its immediate predecessors must be finished:
  - If an activity has only a single immediate predecessor, its ES equals the EF of the predecessor.
  - If an activity has multiple immediate predecessors, its ES is the maximum of all EF values of its predecessors.

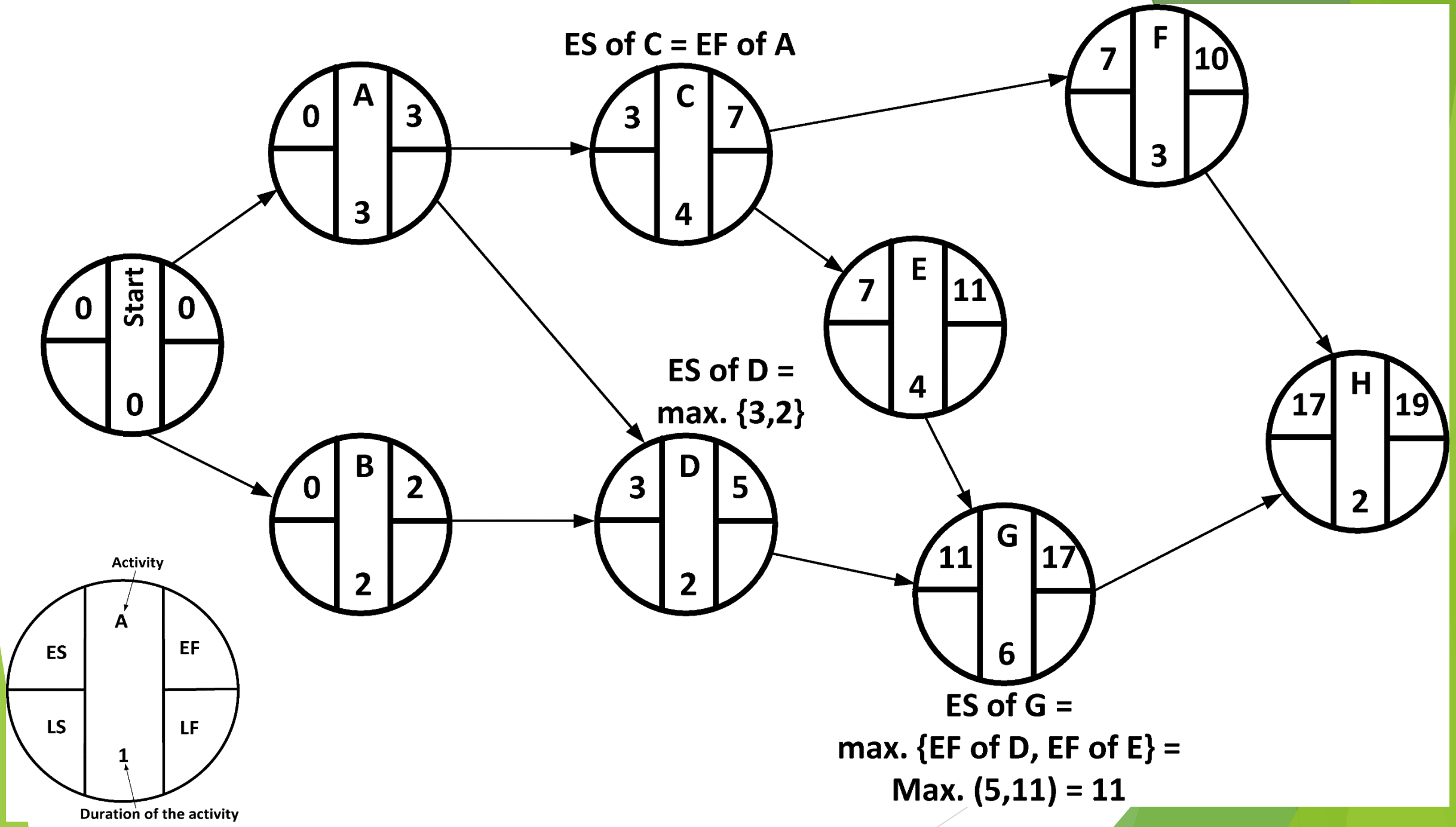
$$ES = Max\{EF\ of\ all\ immediate\ predessors\}$$

Forward Pass

- **Earliest Finish Time Rule**
  - The earliest finish time (EF) of an activity is the sum of its earliest start time (ES) and its activity time.

$$EF = ES + \text{Activity time}$$

- Forward pass allows to determine the earliest project completion time, but it does not identify the critical path.
- To identify this path, backward pass should be conducted to determine the LS and LF values for all activities.


ES of C = EF of A
Start
0
0
0
A
3
0
3
C
7
3
4
F
10
7
3
E
11
7
4
ES of D =
max. {3,2}
B
2
0
2
D
5
3
2
G
17
11
6
H
19
17
2
ES of G =
max. {EF of D, EF of E} =
Max. (5,11) = 11
Activity
A
ES
EF
LS
LF
1
Duration of the activity

Backward Pass

- Backward Pass begins with the last activity of the project.
- For each activity, its LF value is determined first, followed by its LS value. Two rules are used in this process:
- **Latest Finish Time Rule:** This rule is again based on the fact that before an activity can start, all its immediate predecessors must be finished:
  - If an activity is an immediate predecessor for just a single activity, its LF equals the LS of the activity that immediately follows it.
  - If an activity is an immediate predecessor to more than one activity, its LF is the minimum of all LS values of all activities that immediately follow it:

$$LF = Min\{LS \text{ of all immediate following activities}\}$$

Backward Pass

- **Latest Start Time Rule:** The latest start time (LS) of an activity is the difference of its latest finish time (LF) and its activity time.

$$LS = LF - \text{Activity time}$$

LF of C = Min {LS of E, LS of F}
= Min. {7, 14} = 7

